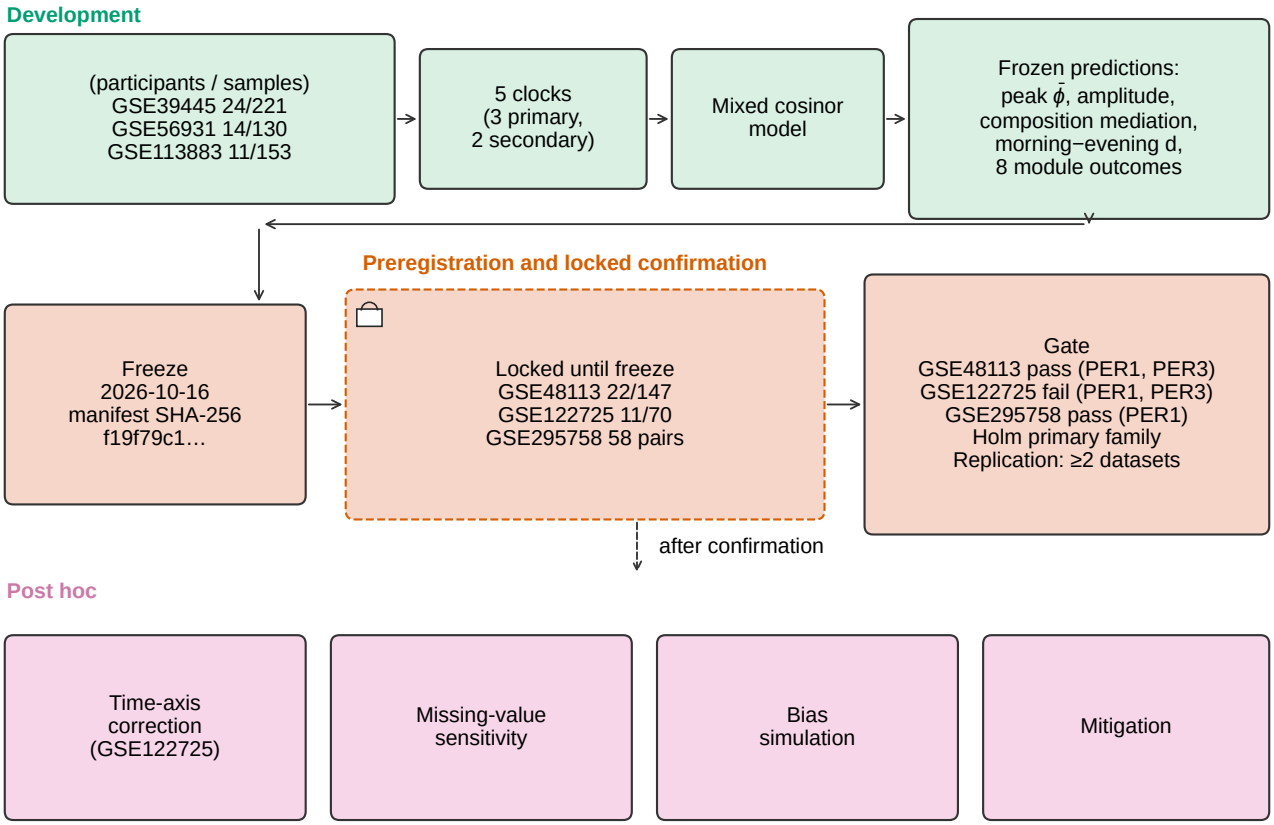
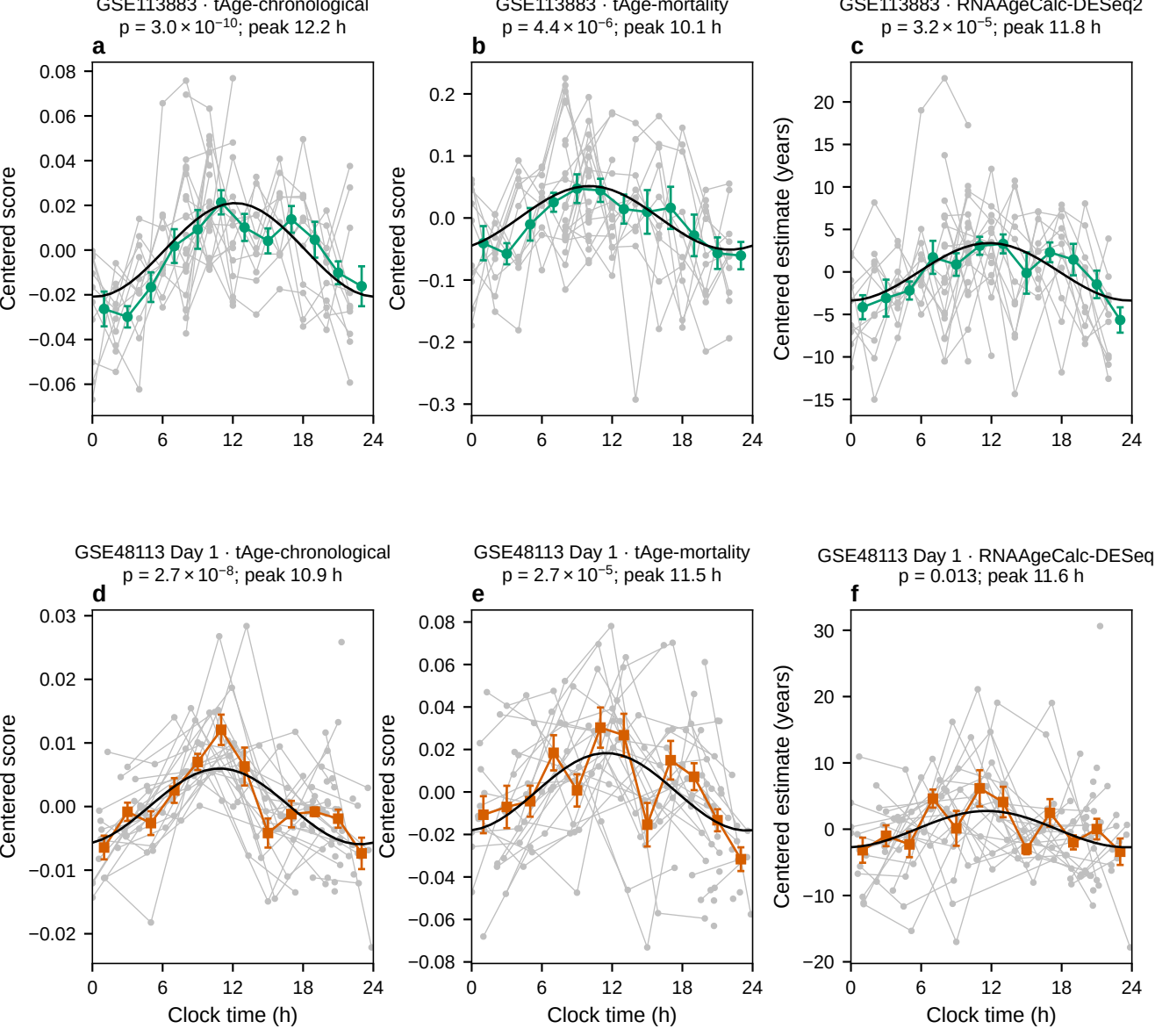
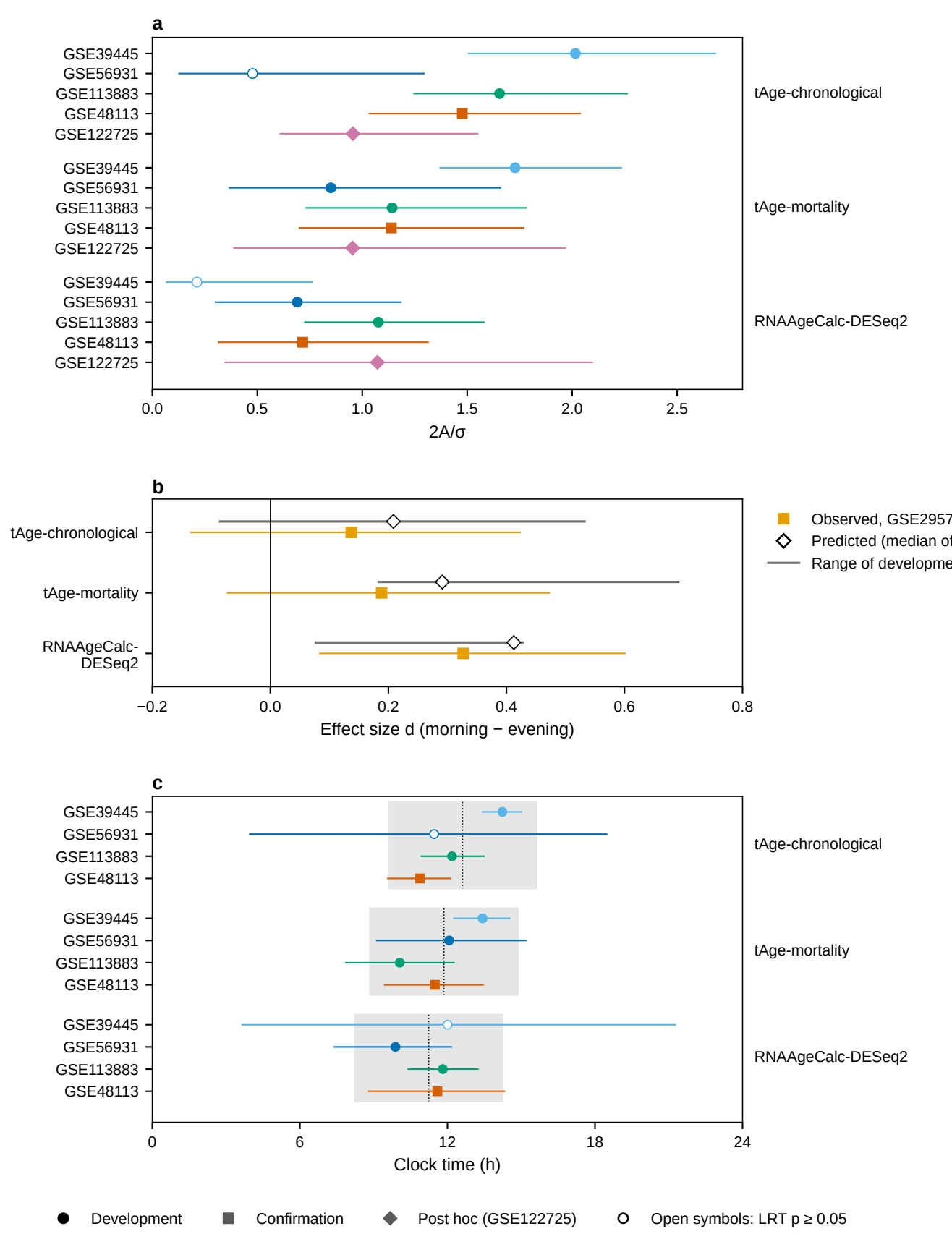
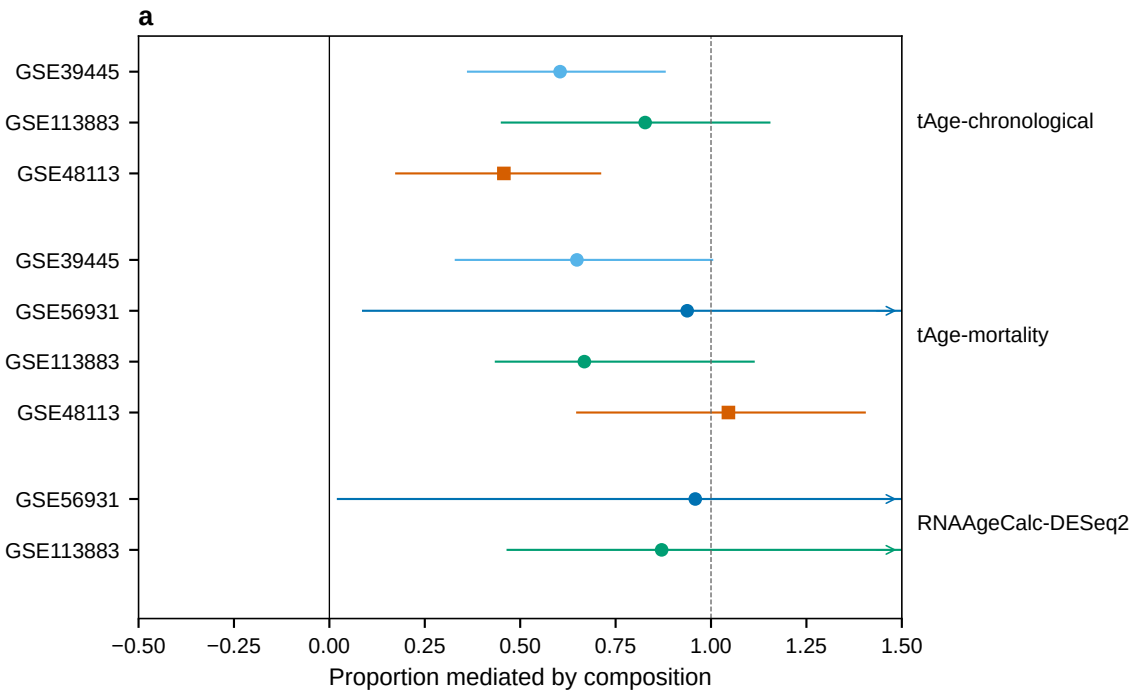
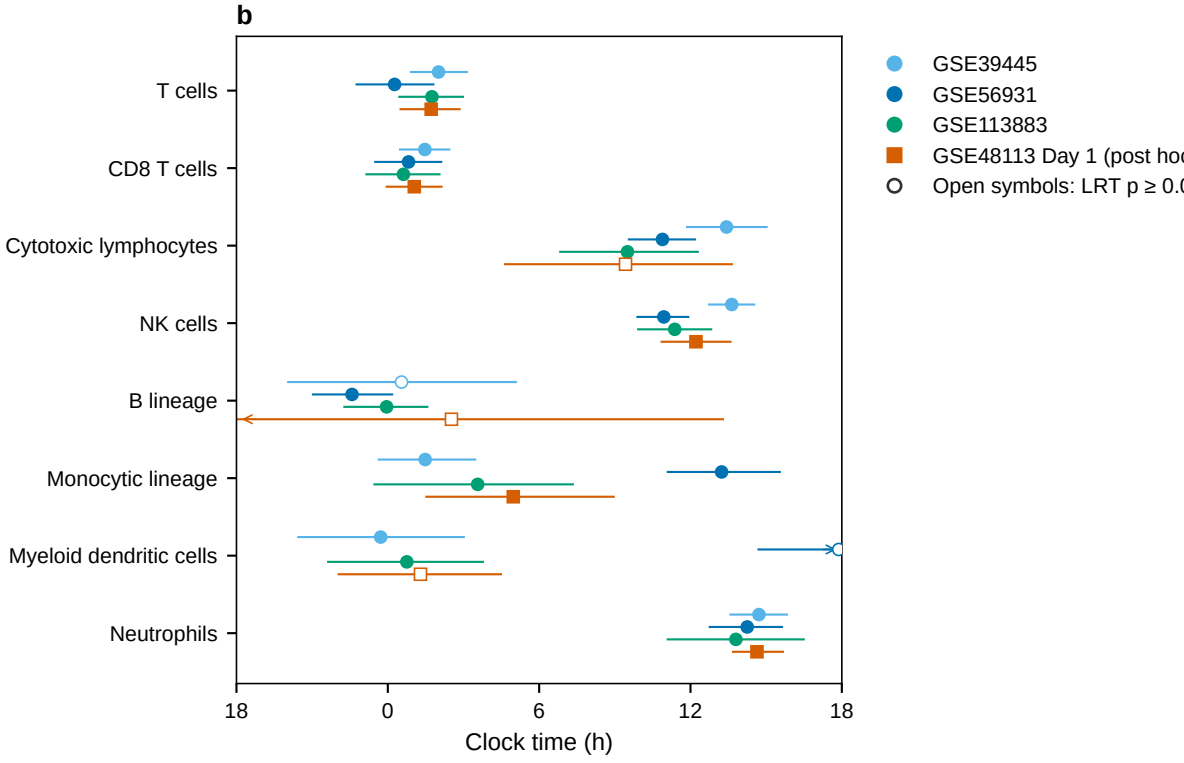
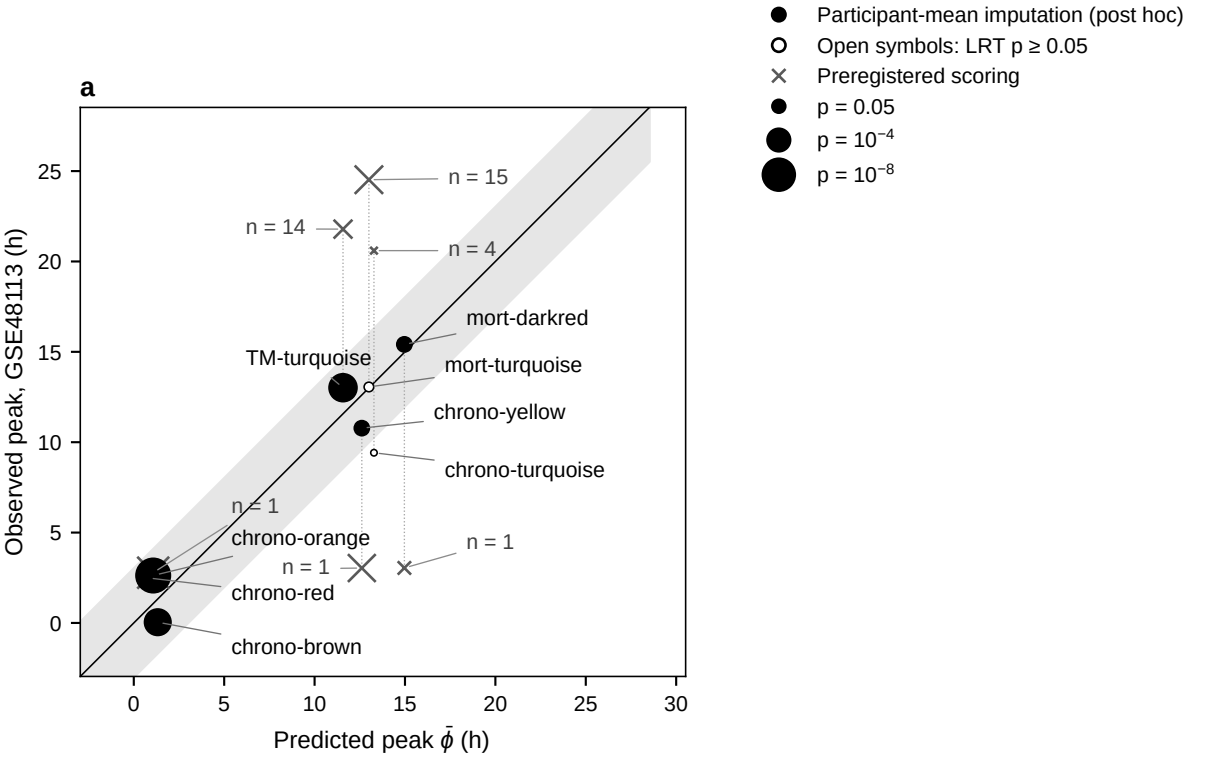

Development
(participants / samples)
GSE39445 24/221
GSE56931 14/130
GSE113883 11/153
5 clocks
(3 primary,
2 secondary)
Mixed cosinor
model
Frozen predictions:
peak $\bar{\phi}$, amplitude,
composition mediation,
morning−evening d,
8 module outcomes
Preregistration and locked confirmation
Freeze
2026-10-16
manifest SHA-256
f19f79c1…
Locked until freeze
GSE48113 22/147
GSE122725 11/70
GSE295758 58 pairs
Gate
GSE48113 pass (PER1, PER3)
GSE122725 fail (PER1, PER3)
GSE295758 pass (PER1)
Holm primary family
Replication: ≥2 datasets
after confirmation
Post hoc
Time-axis
correction
(GSE122725)
Missing-value
sensitivity
Bias
simulation
Mitigation


GSE113883 · tAge-chronological
p = 3.0 × 10⁻¹⁰; peak 12.2 h
a
GSE113883 · tAge-mortality
p = 4.4 × 10⁻⁶; peak 10.1 h
b
GSE113883 · RNAAgeCalc-DESeq2
p = 3.2 × 10⁻⁵; peak 11.8 h
c
GSE48113 Day 1 · tAge-chronological
p = 2.7 × 10⁻⁸; peak 10.9 h
d
GSE48113 Day 1 · tAge-mortality
p = 2.7 × 10⁻⁵; peak 11.5 h
e
GSE48113 Day 1 · RNAAgeCalc-DESeq
p = 0.013; peak 11.6 h
f
Centered score
Centered estimate (years)
Clock time (h)


a
GSE39445
GSE56931
GSE113883
GSE48113
GSE122725
tAge-chronological
tAge-mortality
RNAAgeCalc-DESeq2
0.0
0.5
1.0
1.5
2.0
2.5
2A/σ
b
tAge-chronological
tAge-mortality
RNAAgeCalc-DESeq2
Observed, GSE2957
Predicted (median of
Range of developme
−0.2
0.0
0.2
0.4
0.6
0.8
Effect size d (morning − evening)
c
tAge-chronological
tAge-mortality
RNAAgeCalc-DESeq2
0
6
12
18
24
Clock time (h)
Development
Confirmation
Post hoc (GSE122725)
Open symbols: LRT p ≥ 0.05

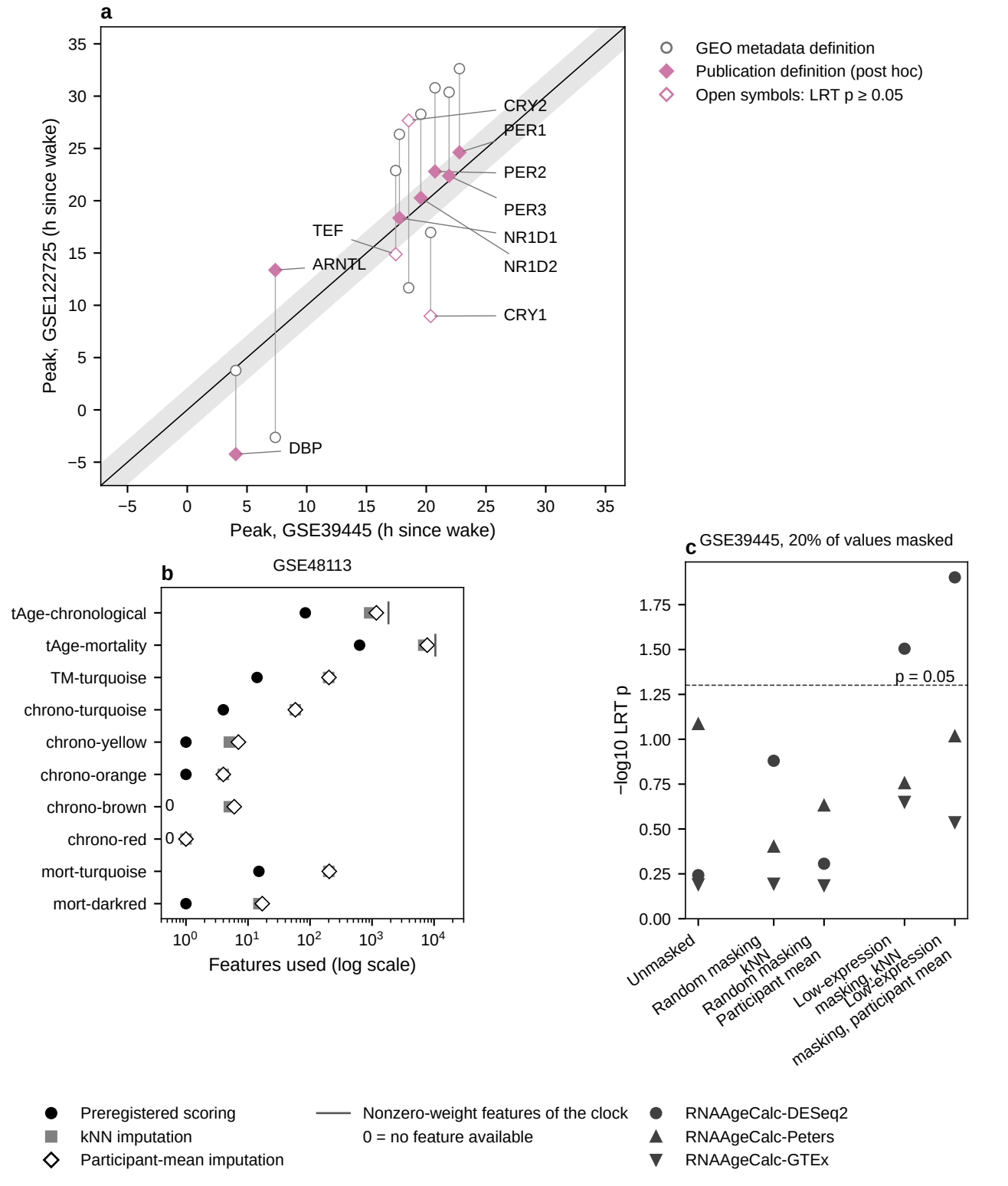

a
GSE39445
GSE113883
GSE48113
GSE39445
GSE56931
GSE113883
GSE48113
GSE56931
GSE113883
tAge-chronological
tAge-mortality
RNAAgeCalc-DESeq2
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
1.25
1.50
Proportion mediated by composition

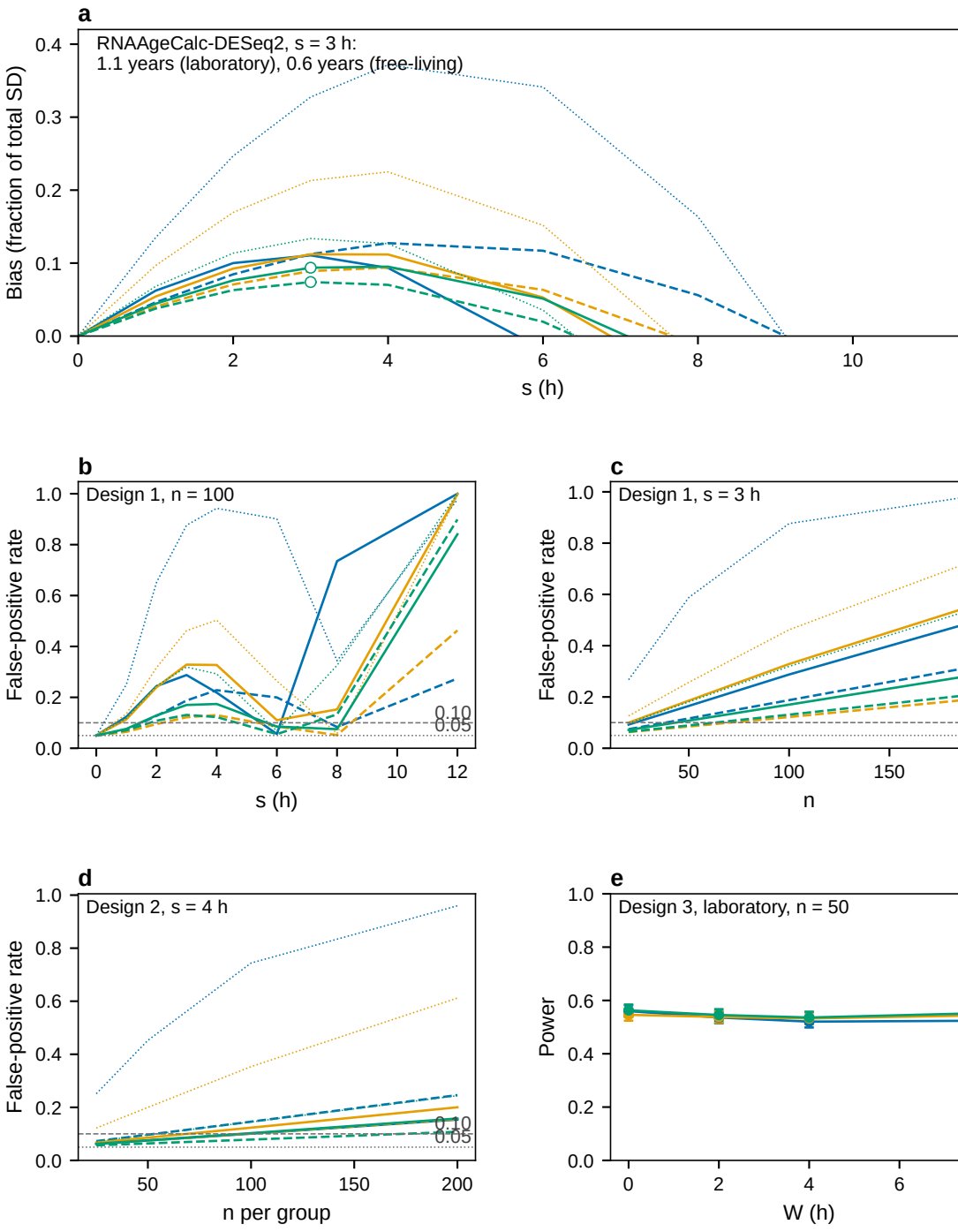

b
T cells
CD8 T cells
Cytotoxic lymphocytes
NK cells
B lineage
Monocytic lineage
Myeloid dendritic cells
Neutrophils
18
0
6
12
18
Clock time (h)
GSE39445
GSE56931
GSE113883
GSE48113 Day 1 (post hoc
Open symbols: LRT p ≥ 0.0

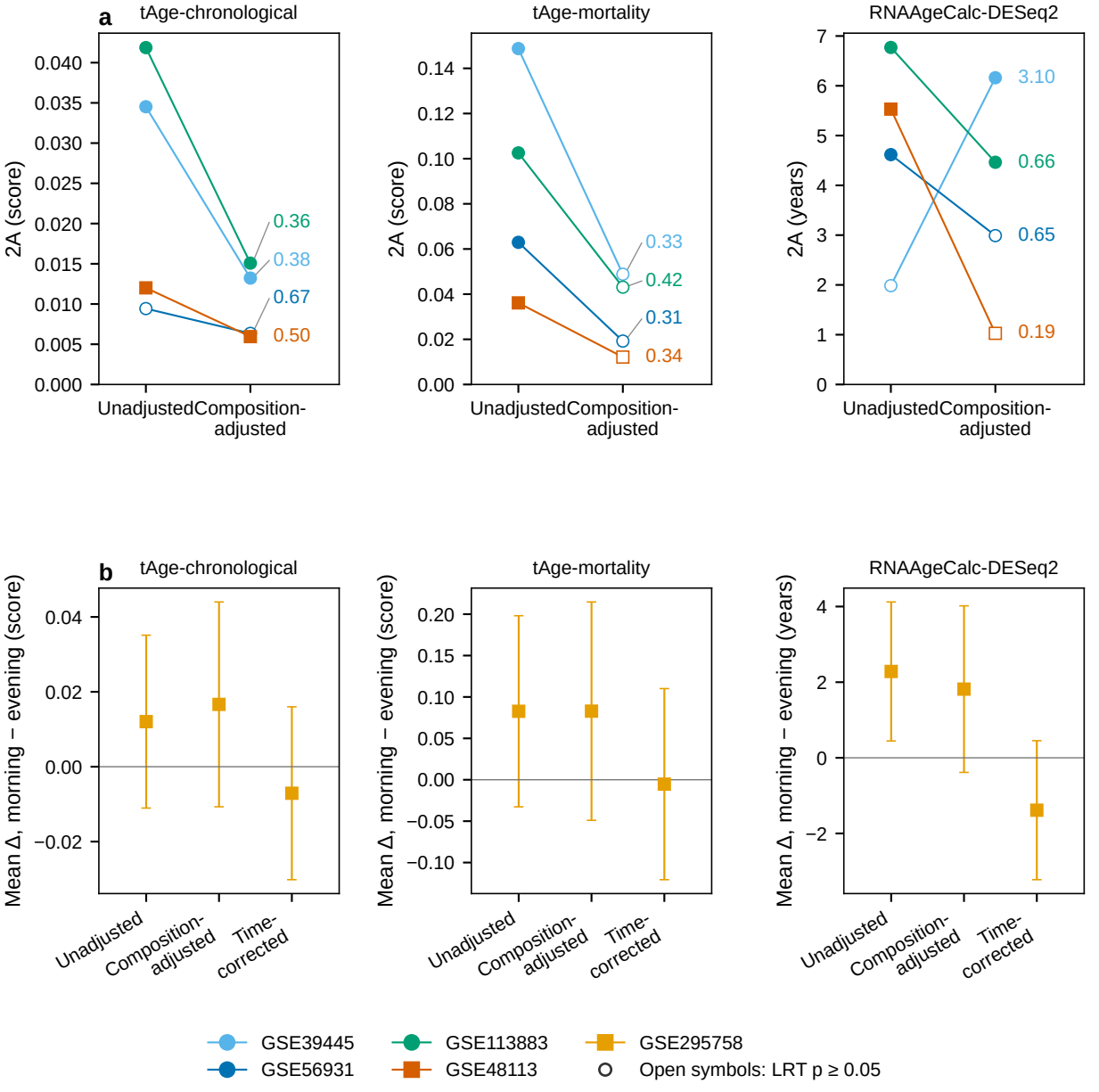


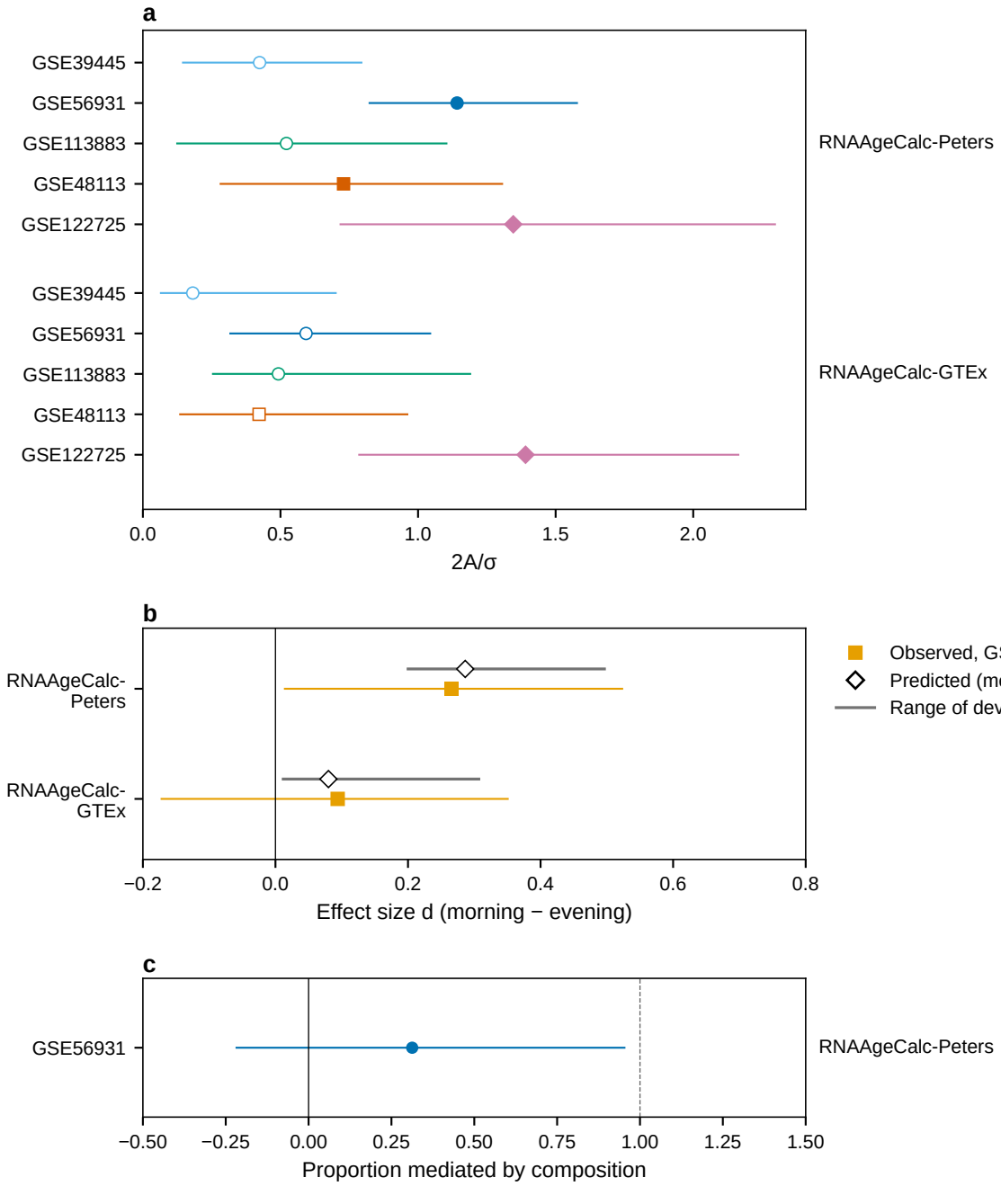
b

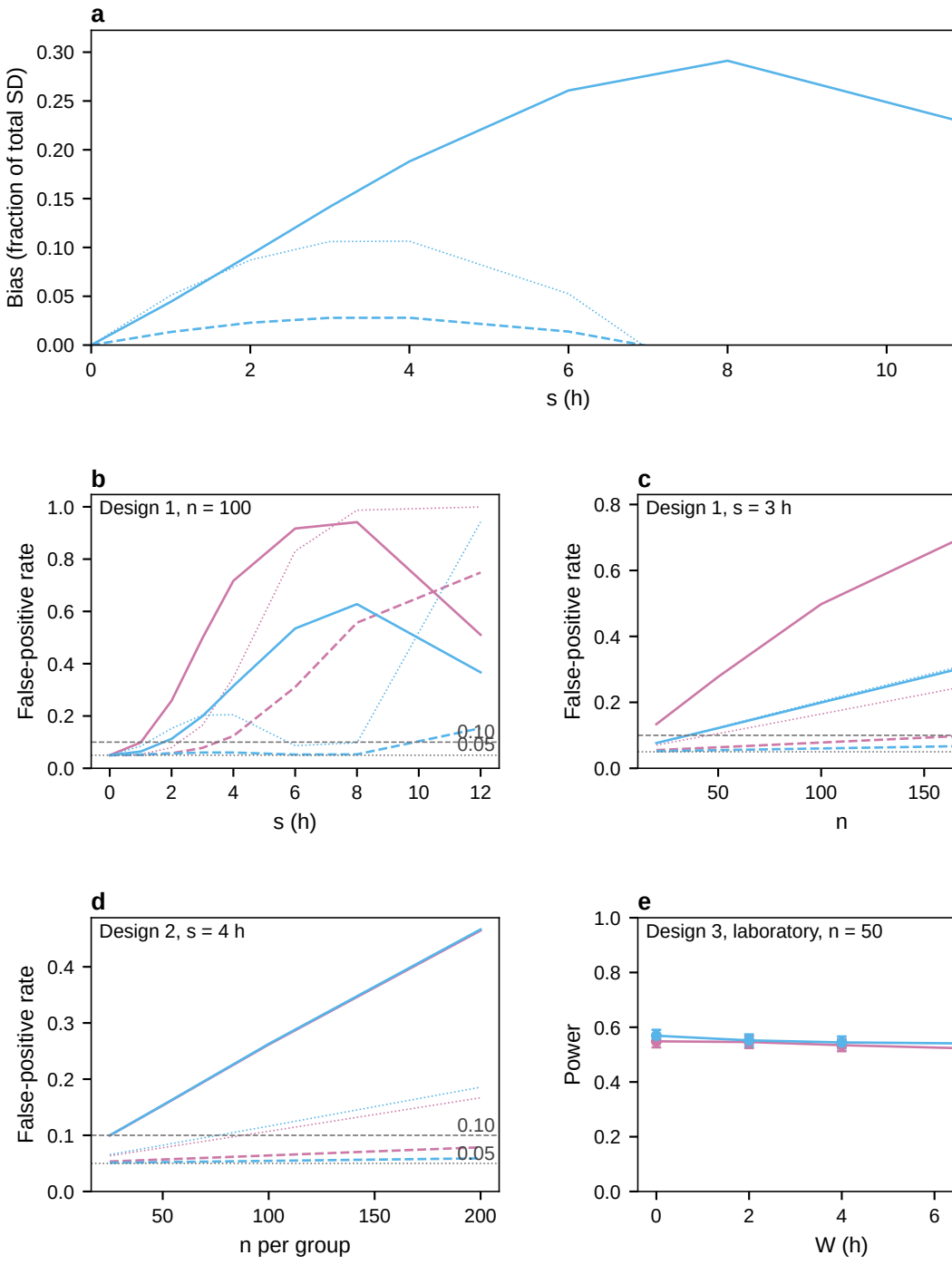
	Development −log10 q			Preregistered scoring		Participant-mean imputation (post hoc)		GSE295758		
	GSE 39445	GSE 56931	GSE 113883	Features	Confirmed	Features	Confirmed	Predicted	Observed	Agree
TM-turquoise	6.8	2.2	5.4	14	No	202	Yes	+	+	Yes
chrono-turquoise	7.6	2.2	2.0	4	No	58	No	+	+	Yes
chrono-yellow	2.8	7.3	8.0	1	No	7	Yes	+	−	—
chrono-orange	8.4	2.2	5.0	1	Yes	4	Yes	−	−	Yes
chrono-brown	7.6	1.4	4.6	—	No	6	Yes	−	−	—
chrono-red	5.2	2.7	2.9	—	No	1	Yes	−	+	—
mort-turquoise	8.2	3.9	6.4	15	No	204	No	+	+	—
mort-darkred	2.5	1.6	7.0	1	No	17	Yes	−	+	—

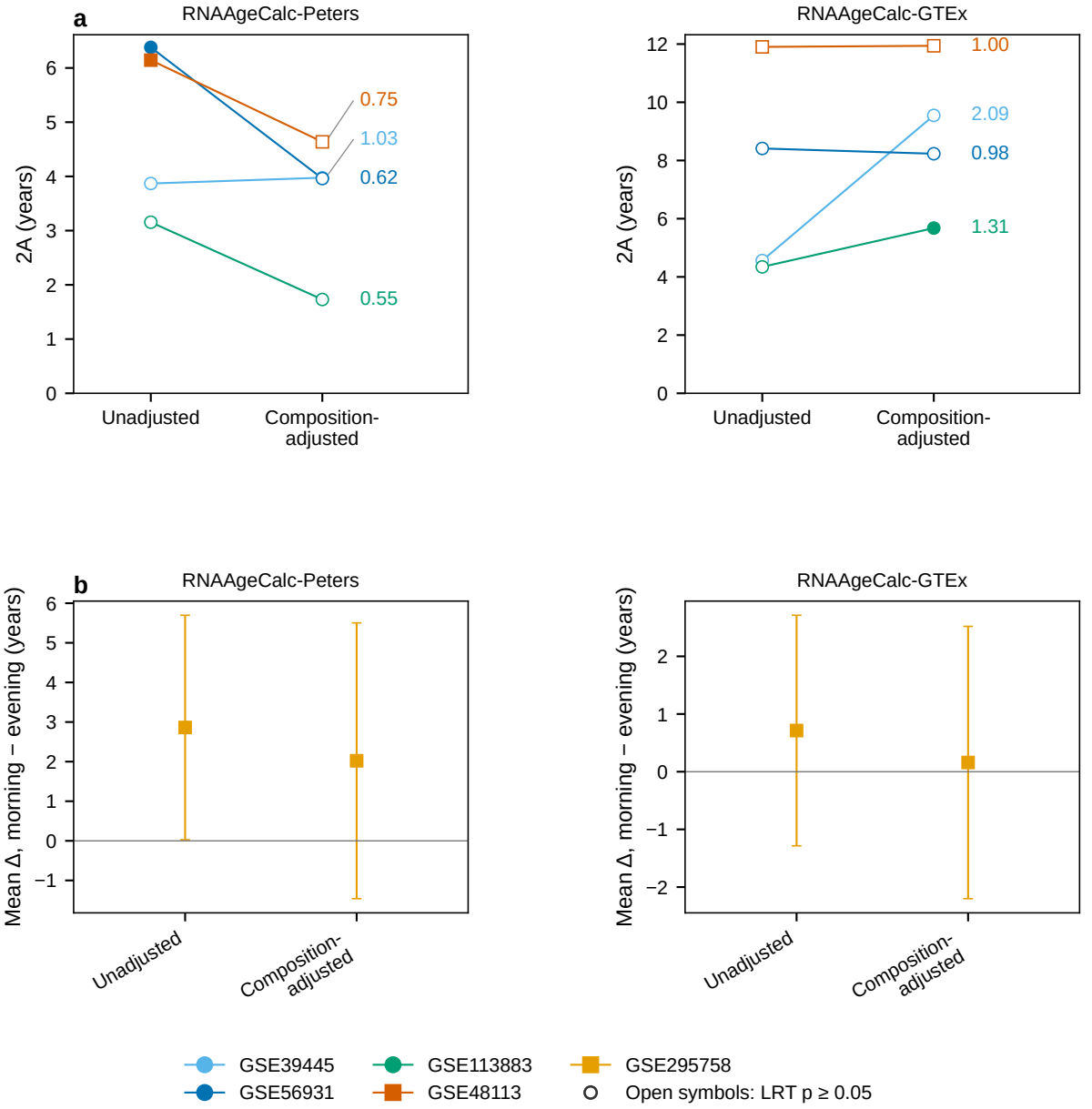

a
GEO metadata definition
Publication definition (post hoc)
Open symbols: LRT p ≥ 0.05
Peak, GSE122725 (h since wake)
Peak, GSE39445 (h since wake)
CRY2
PER1
PER2
PER3
NR1D1
NR1D2
CRY1
TEF
ARNTL
DBP
b
GSE48113
tAge-chronological
tAge-mortality
TM-turquoise
chrono-turquoise
chrono-yellow
chrono-orange
chrono-brown
chrono-red
mort-turquoise
mort-darkred
Features used (log scale)
c
GSE39445, 20% of values masked
−log10 LRT p
p = 0.05
Unmasked
Random masking
kNN
Random masking
Participant mean
Low-expression masking, kNN
Low-expression masking, participant mean
Preregistered scoring
kNN imputation
Participant-mean imputation
Nonzero-weight features of the clock
0 = no feature available
RNAAgeCalc-DESeq2
RNAAgeCalc-Peters
RNAAgeCalc-GTEx


a
RNAAgeCalc-DESeq2, s = 3 h:
1.1 years (laboratory), 0.6 years (free-living)
Bias (fraction of total SD)
s (h)
b
Design 1, n = 100
False-positive rate
s (h)
0.10
0.05
c
Design 1, s = 3 h
False-positive rate
n
0.10
0.05
d
Design 2, s = 4 h
False-positive rate
n per group
0.10
0.05
e
Design 3, laboratory, n = 50
Power
W (h)
tAge-chronological
tAge-mortality
RNAAgeCalc-DESeq2
Laboratory
Free-living
Free-living, upper bound


a
tAge-chronological
tAge-mortality
RNAAgeCalc-DESeq2
2A (score)
2A (years)
0.36
0.38
0.67
0.50
0.33
0.42
0.31
0.34
3.10
0.66
0.65
0.19
Unadjusted
Composition-adjusted
b
Mean Δ, morning − evening (score)
Mean Δ, morning − evening (years)
Unadjusted
Composition-adjusted
Time-corrected
GSE39445
GSE56931
GSE113883
GSE48113
GSE295758
Open symbols: LRT p ≥ 0.05


a
GSE39445
GSE56931
GSE113883
GSE48113
GSE122725
GSE39445
GSE56931
GSE113883
GSE48113
GSE122725
RNAAgeCalc-Peters
RNAAgeCalc-GTEx
0.0
0.5
1.0
1.5
2.0
2A/σ
b
RNAAgeCalc-Peters
RNAAgeCalc-GTEx
−0.2
0.0
0.2
0.4
0.6
0.8
Effect size d (morning − evening)
Observed, GSE295758 (95
Predicted (median of deve
Range of development pre
c
GSE56931
RNAAgeCalc-Peters
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
1.25
1.50
Proportion mediated by composition
Development
Confirmation
Post hoc (GSE122725)
Open symbols: LRT p ≥ 0.05


a
Bias (fraction of total SD)
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0
2
4
6
8
10
12
s (h)
b
Design 1, n = 100
False-positive rate
0.0
0.2
0.4
0.6
0.8
1.0
0
2
4
6
8
10
12
s (h)
0.10
0.05
c
Design 1, s = 3 h
False-positive rate
0.0
0.2
0.4
0.6
0.8
50
100
150
200
n
0.10
0.05
d
Design 2, s = 4 h
False-positive rate
0.0
0.1
0.2
0.3
0.4
50
100
150
200
n per group
0.10
0.05
e
Design 3, laboratory, n = 50
Power
0.0
0.2
0.4
0.6
0.8
1.0
0
2
4
6
8
W (h)
RNAAgeCalc-Peters
RNAAgeCalc-GTEx
Laboratory
Free-living
Free-living, upper bound


a
RNAAgeCalc-Peters
2A (years)
0.75
1.03
0.62
0.55
Unadjusted
Composition-adjusted
RNAAgeCalc-GTEx
1.00
2.09
0.98
1.31
b
Mean Δ, morning − evening (years)
GSE39445
GSE56931
GSE113883
GSE48113
GSE295758
Open symbols: LRT p ≥ 0.05